Elementary Number Theory

AshaIy

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Source Code & Contributions
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Contents

Preface iv

1 Divisibility Theory in the Integers 1
- 1.1 The Division Algorithm . . . 1
- 1.2 The Greatest Common Divisor . . . 1
- 1.3 The Euclidean Algorithm . . . 1
- 1.4 The Diophantine Equation $ax+by=c$ . . . 1

2 Primes and Their Distribution 2
- 2.1 The Fundamental Theorem of Arithmetic . . . 2
- 2.2 The Sieve of Eratosthenes . . . 2
- 2.3 The Goldbach Conjecture . . . 2

3 The Theory of Congruences 3
- 3.1 Basic Properties of Congruence . . . 3
- 3.2 Binary and Decimal Representations of Integers . . . 3
- 3.3 Linear Congruences and the Chinese Remainder Theorem . . . 3

4 Fermat's Theorem 4
- 4.1 Fermat's Little Theorem and Pseudoprimes . . . 4
- 4.2 Wilson's theorem . . . 4

5 Number-Theoretic Functions 5
- 5.1 The Sum and Number of Divisors . . . 5
- 5.2 The Möbius Inversion Formula . . . 5
- 5.3 The Greatest Integer Function . . . 5

6 Euler's Generalization of Fermat's Theorem 6
- 6.1 Euler's Phi-Function . . . 6
- 6.2 Euler's Theorem . . . 6
- 6.3 Some Properties of the Phi-Function . . . 6

7 Primitive Roots and Indices **7**
7.1 The Order of an Integer Modulo n . . . 7
7.2 Primitive Roots for Primes . . . 7
7.3 Composite Numbers Having Primitive Roots . . . 7
7.4 The Theory of Indices . . . 7

8 The Quadratic Reciprocity Law **8**
8.1 Euler's Criterion . . . 8
8.2 The Legendre Symbol and Its Properties . . . 8
8.3 Quadratic Reciprocity . . . 8
8.4 Quadratic Congruences with Composite Moduli . . . 8

9 Certain Nonlinear Diophantine Equations **9**
9.1 The Equation $x^2+y^2=z^2$ . . . 9
9.2 Fermat's Last Theorem . . . 9

10 Representation of Integers as Sums of Squares **10**
10.1 Sums of Two Squares . . . 10
10.2 Sums of More Than Two Squares . . . 10

11 Fibonacci Numbers **11**
11.1 The Fibonacci Sequence . . . 11
11.2 Certain Identities Involving Fibonacci Numbers . . . 11

12 Continued Fractions **12**
12.1 Finite Continued Fractions . . . 12
12.2 Infinite Continued Fractions . . . 12
12.3 Farey Fractions . . . 12
12.4 Pell's Equation . . . 12

13 Some Modern Developments **13**
13.1 Primality Testing and Factorization . . . 13
13.2 An Application to Factoring: Remote Coin Flipping . . . 13
13.3 The Prime Number Theorem and Zeta Function . . . 13

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.34.0 and Mathlib v4.34.0 (revision `5ed296525643`).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

Chapter 1

Divisibility Theory in the Integers

1.1 The Division Algorithm

1.2 The Greatest Common Divisor

1.3 The Euclidean Algorithm

1.4 The Diophantine Equation $ax+by=c$

Chapter 2

Primes and Their Distribution

2.1 The Fundamental Theorem of Arithmetic

2.2 The Sieve of Eratosthenes

2.3 The Goldbach Conjecture

Chapter 3

The Theory of Congruences

3.1 Basic Properties of Congruence

3.2 Binary and Decimal Representations of Integers

3.3 Linear Congruences and the Chinese Remainder Theorem

Chapter 4

Fermat's Theorem

4.1 Fermat's Little Theorem and Pseudoprimes

4.2 Wilson's theorem

Chapter 5

Number-Theoretic Functions

5.1 The Sum and Number of Divisors

5.2 The Möbius Inversion Formula

5.3 The Greatest Integer Function

Chapter 6

Euler's Generalization of Fermat's Theorem

6.1 Euler's Phi-Function

6.2 Euler's Theorem

6.3 Some Properties of the Phi-Function

Chapter 7

Primitive Roots and Indices

7.1 The Order of an Integer Modulo n

7.2 Primitive Roots for Primes

7.3 Composite Numbers Having Primitive Roots

7.4 The Theory of Indices

Chapter 8

The Quadratic Reciprocity Law

8.1 Euler's Criterion

8.2 The Legendre Symbol and Its Properties

8.3 Quadratic Reciprocity

8.4 Quadratic Congruences with Composite Moduli

Chapter 9

Certain Nonlinear Diophantine Equations

9.1 The Equation $x^2+y^2=z^2$

9.2 Fermat's Last Theorem

Chapter 10

Representation of Integers as Sums of Squares

10.1 Sums of Two Squares

10.2 Sums of More Than Two Squares

Chapter 11

Fibonacci Numbers

11.1 The Fibonacci Sequence

11.2 Certain Identities Involving Fibonacci Numbers

Chapter 12

Continued Fractions

12.1 Finite Continued Fractions

12.2 Infinite Continued Fractions

12.3 Farey Fractions

12.4 Pell's Equation

Chapter 13

Some Modern Developments

13.1 Primality Testing and Factorization

13.2 An Application to Factoring: Remote Coin Flipping

13.3 The Prime Number Theorem and Zeta Function

Bibliography

[1] David M. Burton, *Elementary Number Theory*, McGraw-Hill International Editions.